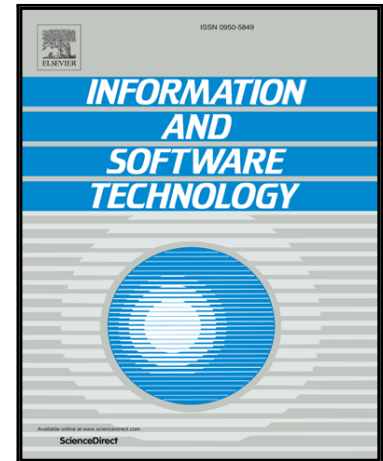


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Evaluation of simulation-assisted value stream mapping for software product development: two industrial cases

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Abstract

Context: Value Stream Mapping (VSM) as a tool for lean development has led to significant improvements in different industries. In a few studies, it has been successfully applied in a software engineering context. However, some shortcomings have been observed in particular failing to capture the dynamic nature of the software process to evaluate improvements i.e. such improvements and target values are based on idealistic situations.

Objective: To overcome the shortcomings of VSM by combining it with software process simulation modeling, and to provide reflections on the process of conducting VSM with simulation.

Method: Using case study research, VSM was used for two products at Ericsson AB, Sweden. Ten workshops were conducted in this regard. Simulation in this study was used as a tool to support discussions instead of as a prediction tool. The results have been evaluated from the perspective of the participating practitioners, an external observer, and reflections of the researchers conducting the simulation that was elicited by the external observer.

Results: Significant constraints hindering the product development from reaching the stated improvement goals for shorter lead time were identified. The use of simulation was particularly helpful in having more insightful discussions and to challenge assumptions about the likely impact of improvements. However, simulation results alone were found insufficient to emphasize the importance of reducing waiting times and variations in the process.

Conclusion: The framework to assist VSM with simulation presented in this study was successfully applied in two cases. The involvement of various stakeholders, consensus building steps, emphasis on flow (through waiting time and variance analysis) and the use of simulation proposed in the framework led to *realistic* improvements with a high *likelihood* of implementation.

Keywords: Case study research, value stream mapping, lean software development, software process simulation modeling, system dynamics

1. Introduction

Enthralled by the success of lean development in manufacturing and product development [1] and the service industry [2], lean concepts have been adapted to the area of software development [3]. One of the practices in lean software development is value stream mapping (VSM) [3]. VSM is the practice of creating a value stream map that identifies the value added by each step in the software development process [4].

Starting with the *Current State Map* (CSM), this practice identifies *value-adding*, *required non-value adding*, and *non-value adding* activities in the current process.

Making this distinction enables the improvement in the current process by the elimination of non-value adding activities [4] and creates a desired *Future State Map* (FSM) and an action plan for improvements.

VSM is a practice that “*straddles the gray area between*” concrete practices and analytical principles of *Lean* [5]. It implements several of Lean principles directly (like “optimize the whole”, “eliminate waste”) and contributes to fulfilling many others (like “continuous improvement”, “flow” and “pull-based” development) [6] [7]. For example, VSM puts to use the principle “*optimize the whole*” by taking several measures to take a system-wide perspective. It does so by consider-

ing the end-to-end process and involving multiple stakeholders responsible for various activities in the process [8], both in identification of waste and improvements. This ensures that the focus is on the most significant impediments hindering the organization from delivering value to the customer.

Moreover, as our understanding of customer value improves we can perform VSM to realign the development process to continue to deliver value efficiently to the customer. This provides the means necessary for “continuous improvement”. VSM helps to identify waste by visualizing and analyzing the current value stream. This helps to operationalize another principle of *Lean* that is to “eliminate waste”. The types of waste (e.g. partially done work, unnecessary features) listed by Poppendieck and Poppendieck [6] should only be considered as a guideline however, it is the VSM activity that will help to see where these manifest in the concrete case of a company’s process and which types of waste should be handled with priority.

Figure 1, depicts how VSM can systematically operationalize the Lean principles by using them first to guide the analysis of the current value stream, then in identification of waste and lastly in identifying which improvements to implement.

The use of VSM has led to several tangible benefits e.g. McManus [9] reports studies where organizations significantly reduced the standard deviation of the cycle times, the rework required to fix defects and the cycle time by 75%. It is also reported that typically up to 25% savings in engineering efforts are achieved with VSM based improvements [9].

Similar to manufacturing processes, in the development process for aerospace, information inventories had “engineering work-packages” inactive for up to 77% of the time [9]. The success of VSM outside manufacturing and production and in the design and development process of engineering makes it interesting to explore if it can be leveraged in software development.

Khurum et al. [8] adapted and applied VSM in the context of large-scale software intensive product development. They found the following two shortcomings based on practitioners feedback [8]: First, the notation used only provides a snapshot of the system and fails to capture the dynamic aspects of the underlying processes. This leads to a simplistic analysis to identify bottlenecks. Second, the improvement actions and the target value maps are based on idealistic situations. This also reduces the realism in the target value stream map (as it is based on ideal conditions that the identified problems are completely resolved).

These limitations reduce the confidence in the improvement actions identified in VSM, making it less likely that such improvement actions will be implemented. Given these limitations [8], and the potential benefits of software process simulation modeling (SPSM) [10], we argue that SPSM can be used to overcome these shortcomings. Other disciplines where VSM has been used have also seen a utility in combining VSM with simulation (cf. [11, 12, 13, 14, 15, 16]). Some reasons for such combination in these disciplines are listed below:

- Simulation helps to reason about changing the process, and supports consensus building by visualizing dynamic views of the process [11, 12, 17, 18, 19, 20, 21, 22, 23, 24, 25]
- VSM alone is time consuming and simulation can assist to speed-up the analysis of the CSM, and the derivation and verification of a FSM [25, 26, 27, 28]. Simulation helps to verify the impact of changes and answering questions that cannot be answered by VSM [13, 28, 29, 30]
- VSM provides a snapshot, it is unable to detail dynamic behavior. Simulation can help predict the flow and levels and provide more accurate quantitative measures given its ability to handle both deterministic and stochastic inputs [16, 17, 26, 31]
- VSM alone cannot capture the complexity in terms of the iterations, overlaps, feedback, rework, uncertainty and stochasticity of the process [14, 26, 32, 33, 34, 35]

The contributions of this study can be briefly summarized as the following:

Contribution 1: Based on literature, this study proposes a framework to perform simulation-assisted VSM.

Contribution 2: Evaluation of the use of VSM and the simulation support in industry.

To achieve the evaluation contribution, we conducted the study at a development site of Ericsson AB, Sweden. This entailed ten (three hours long) workshops for two products with five to six relevant practitioners in each workshop, and extensive quantitative and qualitative data gathering and analysis.

The evaluation of simulation-assisted VSM was done from three different perspectives, namely: (a) the practitioners participating in the activity, who are to benefit from and utilize the approach (b) the facilitators (first and the second author of the article) who conducted the simulation-assisted VSM (c) an external observer (third author of the paper) who is a simulation expert being

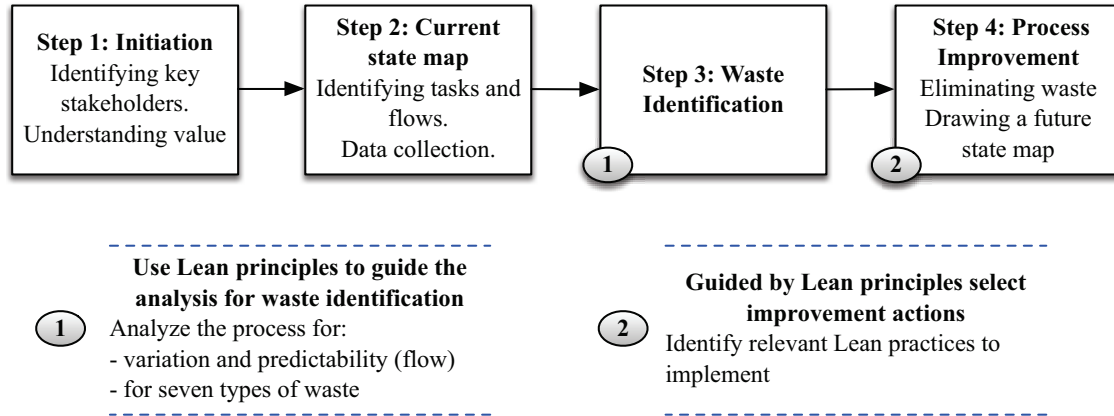


Figure 1: Overview of the VSM steps and their role in the operationalization of Lean principles.

present during the workshops. The external observer assesses the process from an academic point of view. His expertise (having defined guidelines for simulation and prior hands-on simulation experience) allowed a critical observation of our use of simulation in a practical situation. He had a guideline focus compared to the practitioners, facilitators and our view as researchers doing applied research using simulation in the context of VSM.

Furthermore, we provide a detailed account of the findings to illustrate which wastes and improvements could be identified through this approach.

The remainder of the paper is structured as the following: Section 2 presents related work. Section 3 maps the strategies for combining simulation with VSM and details the process for performing it (Contribution 1). Section 4 describes the research method. Application and the results of using simulation-supported VSM (Contribution 2) are presented in Section 5. Section 6 discusses the findings and Section 7 concludes the paper.

2. Related work

Through an extensive search and by considering references that existed in secondary studies on “lean” in the software engineering (SE) context, we identified three primary studies reporting the use of VSM in software development [8, 41, 42].

Yang et al. [32] have applied VSM to improve the lead-time of an IT company’s R&D process for a hardware component. Mujtaba et al. [41] used VSM as a means to reduce the lead-time in the product customization process. Kasoju et al. [42] have used VSM to im-

prove the testing process for a company in the automotive industry. Khurum et al. [8] have extended the definition of waste in the context of software intensive product development. While value should be considered fully from the perspective of the customer, they have added further dimensions potentially relevant for the customer that were traditionally not considered. However, these value considerations should be evaluated on a case to case basis considering what each organization considers important from their customers’ perspective. They also reported the process of conducting a VSM.

Khurum et al. [8], Kasoju et al. [42] and Mujtaba et al. [41] have used conventional static VSM notation. Yang et al. [32] have applied VSM in combination with simulation for a hardware product.

To the best of our knowledge, this is the first study focusing on the investigation of simulation-assisted VSM for software product development. However, from literature in manufacturing, production, service industry and product development five strategies were found to combine VSM with simulation, these are summarized in Table 1. In Section 3, we map these strategies on the process [8] for conducting VSM.

3. Framework for simulation-assisted VSM (Contribution 1)

The challenges that motivated us to explore the use of simulation in conjunction with VSM are very similar to the motivations that researchers from other disciplines have had for this combination (see details in Section 1).

Building on the research from other disciplines, we will map their strategies (see Table 1) to combine simulation with VSM to the VSM process as defined for

ID.	Strategy	Purpose	References
1	Developing simulation model for both the CSM and FSM of the system	Compare the CSM and FSM to demonstrate the benefits achieved by improvements.	[11, 12, 18, 20, 27, 23, 33, 34]
2	Develop simulation model for the FSM	To assess the effectiveness of the proposed improvements in terms of the intended benefits, and the likelihood of achieving those benefits.	[13, 29] [22, 30, 32, 35, 36] [24, 21, 37, 28, 25]
3	Develop a simulation model of the CSM	To identify or verify the bottlenecks in the CSM.	[19]
4	Develop simulation model to derive FSM	To evaluate several alternatives for FSM. To find optimum values for parameters like batch-sizes, work-in-progress, resource allocation and scheduling etc.	[19, 17, 38, 15, 39]
5	Use VSM based notation as a front-end for simulation model	To automatically generate simulation models from the notation used. This approach can achieve all the intended uses of simulation, in combination with VSM, as identified in the above strategies.	[31, 40, 14, 16, 26]

Table 1: Overview of research on the combination of VSM and simulation.

the SE context [8]. Moreover, we provide useful references to SPSM research [43] that will guide the use of these strategies for conducting VSM with simulation for software product development. We also discuss the implications and likelihood of success of these strategies for VSM in the SE context.

To successfully conduct a VSM in an organization the following three roles are required: **VSM Manager:** a person who acts as the champion for the VSM activity [44] having authority to adapt the goal from upper management, get the commitment from participants for availability. **VSM Facilitators:** require a mix of knowledge and experience of lean and product development improvement expertise [9] and they are responsible for moderating the sessions. **VSM Participants:** practitioners with knowledge and experience encompassing the end-to-end process [9].

The VSM process has five distinct steps as shown in Figure 2. Five strategies (see Table 1) to assist VSM with simulation are placed appropriately in the VSM process (as shown in the Figure 2).

Except the first step that is handled in a meeting with a few key participants (as detailed in Section 3.1) all the remaining steps are done in a workshop setting. The workshop enables the participants to have discussions and allows taking a consensus driven approach to the end-to-end process improvement [8].

For more details on each step and some useful templates for data collection please refer to Khurum et al. [8]. A summary of the activities, main purpose, roles involved, data sources used, and types of analysis done are presented in Sections 3.1 to 3.4 along with the additional information on how simulation can support each of these steps.

3.1. Initiation

The first step S1 (see Figure 2) constitutes the prerequisite activities that are required to support and successfully complete a VSM in an applied setting. The focus is on getting support from the relevant stakeholders and to engage the right persons in the activity. Furthermore, the scope of the process to be investigated (start- and end-points) is determined.

3.2. Current state map

In the next step S2 (see Figure 2), an attempt to understand the current value stream is made. This step aims to go beyond the documented process and really make explicit the current way of working. This step is essential to establish a current state map, as in practice often the processes are poorly documented, which may refer to obsolete practices, contain irrelevant details or the details that were once critical but have become irrelevant over time [9].

Although it is not mandatory, VSM typically uses a specialized notation to draw the current map [9, 44]. However, in previous experience [8], it was found that practitioners found it more intuitive to represent the process in the way they perceive and experience it.

Simulation assistance: Literature on VSM outside SE, suggests the use of a notation that can be used to automatically generate simulation model of the current state map (see Strategy 5 in Table 1 and the related literature, such as [31, 40, 14, 16, 26]). Theoretically, this is the most appealing strategy as once this automatic translation is achieved, there is no additional cost of building a simulation model and one gets all the benefits of having a dynamic model. In SPSM, work by Hsueh et al. [45] is very pertinent. They suggest the use of a UML based notation to document the software development and then generate simulation models from it.

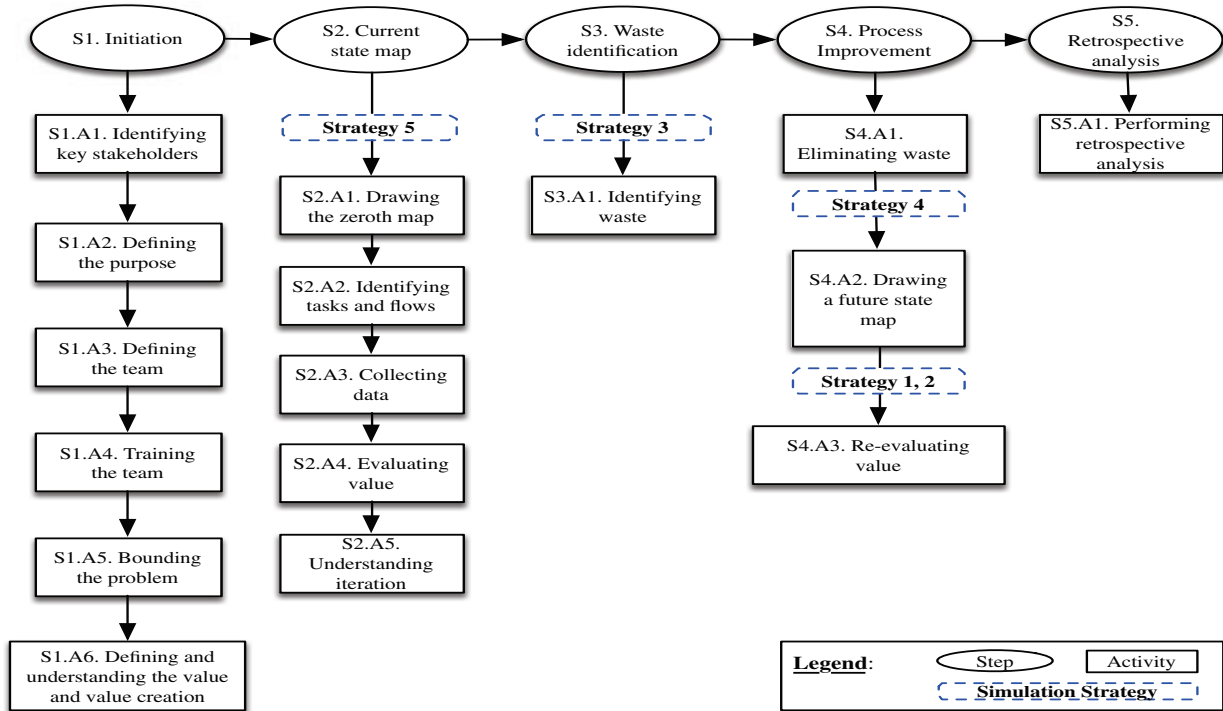


Figure 2: Overview of the VSM process and potential points for the use of simulation.

Once a dynamic model of the process is available, it can be used for all the various purposes as listed in Table 1, e.g. to identify bottlenecks in the current state map. The incoming requests' pattern can be changed to simulate the increase in enhancement requests, mimicking the increased load on the process due to the product's launch in a new market.

However, not many companies may fulfill the prerequisites of having UML models of their processes as a baseline [46]. Thus, this very promising strategy to have process models in UML that can be simulated, may not be practical in a majority of the companies. For others interested to explore this strategy of combining simulation, SPSM research from Hsueh et al. [45] should provide a starting point.

Another limitation of the proposals of using UML to capture the CSM goes against the recommendation of *drawing the value stream map by hand and resisting the use of computers* [44].

3.3. Waste identification

In step S3, data available in the organization or informed estimates are acquired for key parameters in the value stream. The number of work units, artifacts produced, rework, estimates for the size and effort, defect

data, average processing times and variance are typically collected. To facilitate and to improve the reliability of results, the activity of identifying wastes should use a combination of qualitative (of data acquired in the workshops) and quantitative analysis (of empirical and simulation generated data). In particular, quantitative data is required to follow Strategy 3 for the use of simulation.

The definition of lean emphasizes achieving the ability to deliver value in an uninterrupted and predictable way (i.e. continuous flow) when requested by the customer (i.e. pull-driven) [7]. In line with the definition, Rother and Shook identify them among the guidelines for a lean value stream [44]. We briefly describe the two concepts below:

Continuous flow means that work items flow through the process at a continuous pace. A detailed account of definitions of continuous flow and reasons for the lack thereof is presented in [47, 48]. A lack of continuous flow, in combination with the waiting times, is a first indication of waste in the process, and should be further investigated [8].

Pull driven process is an aim for lean, since this ensure that one process produces only what and when the next process needs it, this will help achieve shorter

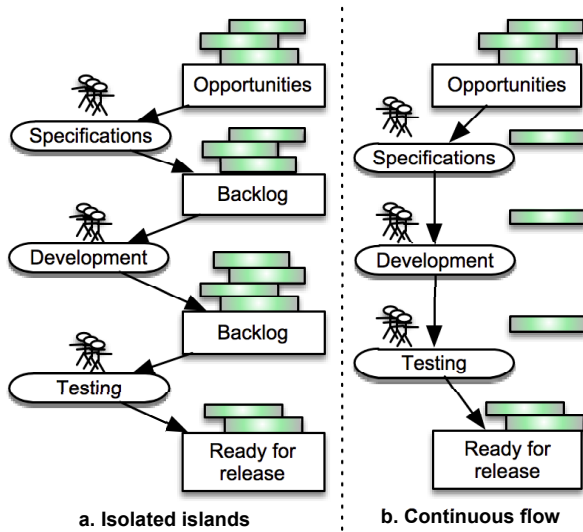


Figure 3: Continuous flow instead of isolated islands based on [44]

lead time and higher quality at a lower cost of development [44]. This has been illustrated in Figure 3. The figure shows that in the lack of “pull” driven development, we observe large backlogs of work in all process activities, also referred to as partially done work [3]. Early in the process this is not an issue as requirements represent ideas that one can draw from, and no large investment has been made. Though, if the company has a large backlog of untested code then an investment has been made in the implementation that does not flow back to the company in terms of revenue. Moreover, when having too long waiting times, already implemented requirements may become obsolete [49].

Simulation assistance: Strategy 3 in Table 1 suggests the use of simulation to identify new bottlenecks or to verify the existing bottlenecks [19].

If the data to accurately simulate the entire CSM is available then an analytical comparison should be sufficient to verify the existence of bottlenecks and developing a simulation model just to verify known bottlenecks will be unnecessary.

However, if there is a change that is going to happen in the future then simulation can be used to identify and verify any potential bottlenecks in the CSM. For example, if we know that in the future we will have to increase the number of maintenance requests we can simulate and see if the current process can handle the increased flow and identify where the bottlenecks will appear (or verify if that concurs with our perception of bottlenecks) or are likely to manifest in the future.

In SPSM, almost any of the works on the use of simulation for strategic management [50], control and operational management ([51]), planning [52, 53], technology adoption and process improvement [54, 55] can be used depending on the scope of the process and particular focus of change that needs to be investigated. A comprehensive coverage of industrial applications of SPSM for a variety of purposes can be found in a study by Ali et al. [43].

3.4. Process improvement

In this step, we revisit the findings from the various analyses and discussions. In some cases, we also present results of lean transformation from other products within the company or from other companies having a similar industrial context. With these inspirations and points to consider, certain changes are proposed to eliminate waste and a future state map is drawn.

Typical questions for arriving at the FSM from the current state map in VSM literature are not entirely applicable in SE [44].

Simulation assistance: Given that quite often we will have more than one way of addressing a challenge in the process (or eliminate a waste), simulation is suggested to evaluate and choose from these alternatives. This is “Strategy 4” in Table 1 as depicted in the Figure 2. Thus, a decision can be made by evaluating the alternatives and then designing the FSM accordingly. Some literature on evaluation of improvement alternatives using SPSM include [55, 56], for a more thorough list c.f. [43].

Khurum et al. [8] recommend to “Re-evaluate value” to assess the impact of changes that have been recommended to eliminate waste. The strategies for the use of simulation to assist in this step are Strategies 1 and 2 (as shown in Table 1).

3.5. Retrospective analysis

In this step, we collect feedback about the process and outcomes of VSM. This helps to identify shortcomings and improve future VSM undertakings.

In this study, Strategy 4 (see Table 1 and Section 3.4), that is to “develop a simulation model to evaluate several alternatives to derive a FSM” was evaluated using the research method described in Section 4.

4. Research methodology

According to Runeson and Höst’s [57] adaptation of case study research, based on the purpose of the study, it can be classified as case study research with the aim of

improvement. The phenomenon under study is “*value stream mapping*” in a software intensive product development context. We explored and evaluated the utility of simulation in supporting the VSM.

4.1. Research questions

Pernstål et al. [58] found in a systematic literature review that while VSM is a “*central practice*” in Lean, only two papers report the use of VSM in the SE context out of a total of 38 included papers. Therefore, due to the scarcity of the reported applications of VSM in the SE context the following research question was posed:

RQ1: *Is the value stream mapping process adapted from Khurum et al. [8] capable of establishing the current state, identifying waste, and developing an improved future state of the process?* We report details of activities and their outcomes to facilitate further applications of VSM. Furthermore, the description of the overall VSM activities and identified challenges help to understand the context for the use of simulation, which is the subject of the second research question.

RQ2: *How useful is the combination of simulation and value stream mapping, and what are its strengths and limitations in that regard in the software engineering context?* The following two sub-questions were addressed to assess the usefulness of VSM supported with simulation:

- *RQ2.1: Which wastes and improvements could be identified with VSM combined with simulation?*
- *RQ2.2: How do practitioners, facilitators and an external simulation expert perceive the process and outcomes for the VSM assisted with simulation?*

These research questions were answered through two case studies involving two different telecommunication products at Ericsson AB.

4.2. Research context

Ericsson AB is a large, ISO 9001:2000 certified telecommunication vendor. The two products studied here have different levels of maturity, type of users and number of customers. Both products have different approaches to development, and are not part of the same overall system structure. That is, their similarities are limited to shared aspects of company culture (e.g. terminology, etc.), the product domain, and the overall high level process, which are depicted in Figure 4. Therefore, in this study they are considered two independent cases.

The process starts by breaking down a *Business Opportunity* (BOP) into high-level *Business Use cases*

(BUCs) that describe the system level functionality which the system must deliver to exploit the opportunity in the market. BUCs take into consideration the product anatomy, technical requirements and the business context and provide a more detailed specification. BUCs are the work-item used to plan, control and manage development progress in the company. A more detailed analysis of BUCs is performed and broken down to user-stories that are implemented by the system development teams.

While both products use agile practices in development, there are structural differences in how the work is organized; e.g. one uses fixed-length sprints and the other uses a variable-length sprint (where the length depends on the estimated time required for the development of a BUC). For both products, a development sprint is followed by a three week testing-sprint that covers system-level testing. The two cases in this research are referred to as product A and B.

Product A: Product A is a relatively new offering from the case organization, but has a large potential market. Design and testing is done in 3-weeks sprints. The main programming language used is Java and GIT is used for source code version management. The following teams are involved in its development located in India and Sweden:

- One team each for system management and system testing. System management is responsible to determine how to integrate the requirements in the form of BUCs into the existing system since they know the system’s anatomy very well.
- Seven teams for development (each with five developers and two testers)

Product B: It is a configuration product that has to interface with various databases and applications. It conceals the complexity of interconnectivity between different vendor software. It is not marketed as an independent product. It has one major release every year. Product development is done in two locations, China and Sweden, where the following teams are involved.

- One team each for system management, architecture support, test automation and system testing
- Four teams for development (each with 5 developers)

4.3. Data collection and analysis

In this section, different means of data collection used to answer the research questions RQ1 and RQ2 are discussed.

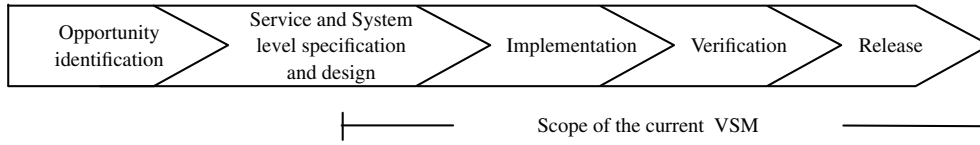


Figure 4: Overview of the product development process in the case company.

4.3.1. Documenting work products of VSM (RQ1)

The primary means for collecting the data during the study were the workshops guided by the VSM process. In order to increase the reliability of the collected data, two researchers were involved. One took notes and the other researcher was mainly responsible for moderating the VSM activity. In addition, we took photographs of all collected materials on the white boards.

We relied on the outcomes of each of the workshop to capture the practitioners' perspective on aspects under study e.g. the current development process as they perceive it, the main challenges which they consider need to be fixed first, what improvements do they propose. In each subsequent workshop the results of our analysis of the previous one were presented to validate them. Furthermore, the triangulation of the qualitative input from the practitioners against archival data from the company was done during these workshops. Thus, the data collected from these workshops helps to address the research questions of the study in particular by identifying wastes and improvements.

Quantitative data regarding the process metrics such as backlog sizes for various phases, time while an employee or a team started working on a BUC, and data to track progress of BUCs through various stages of the development process were extracted from the company database. Effort estimates for the BUCs were also extracted from the archives to perform in-depth analysis. Such data was analyzed using descriptive statistics and visualized in figures which are typical for VSM analysis (see Figures 5, 6 and 7).

A questionnaire (discussed in more detail in Section 4.3.3) was used to collect practitioners' feedback about the usefulness of both the VSM process and the outcome (see questions Q6 to Q9 in Table 2).

4.3.2. Development of the simulation model (RQ2.1)

Alternative scenarios for structuring development and test iteration were elicited in a workshop setting. The data acquired for RQ1 was used to calibrate the simulation model as well.

4.3.3. Reflection and observation of practitioners and an external researcher (RQ2.2)

To elicit feedback from the practitioners about the usefulness of both the process that was followed to conduct simulation-assisted VSM and the outcome of the VSM activity, we conducted a moderated workshop. Semi-structured questions and a structured questionnaire were used to systematically acquire qualitative feedback. Participants from both product A and B filled in the questionnaire (Table 2) anonymously. The coverage of various aspects of this research by questions posed in this questionnaire can be classified as shown in Table 2. The participants of the study filled in the questionnaire during a workshop session which allowed them to ask questions for clarification. In addition, we acquired qualitative feedback through discussion during the workshop. Questions Q1 to Q5 of the questionnaire (see Table 2) in particular attempt to capture the practitioners' perspective on simulation-supported VSM i.e. to answer RQ2.2.

The same questionnaire (see Table 2) along with additional questions about the benefit of the approach was used to assess the usefulness of the approach from the "VSM Manager's" perspective.

An external observer (the 3rd author) was present during the workshops. The external observer used three different approaches to acquire data: participation in the workshops as an observer; interviews with the VSM facilitators; document analysis and informal meetings.

Participation in workshops: The external observer during the workshops took notes about observations related to the outcomes of the VSM activity, as well as what he observed in relation to the use of simulation from the perspective of a simulation expert.

Informal meetings: These unstructured/non-systematic meetings were important to capture explanations and increase the understanding of observed activities, roles, decisions, and contextual information, both from the case under study at Ericsson and the applied methodology.

Interviews: Two separate interviews with the researchers (first two authors of the article) involved in the case study were conducted to capture their opin-

ID	Question	Evaluated aspect			
		Process	Outcome	VSM	VSM + Simulation
Q1	'We have gained new insights about our software development processes which we did not have before.'		X		X
Q2	'Overall more insightful discussions resulted from the simulation results shown.'		X		X
Q3	'The simulation result shown was important to make a convincing point about the importance of reducing variance to achieve improvements.'		X		X
Q4	'The simulation result shown was important to make a convincing point about the importance of reducing waiting times to achieve improvements.'		X		X
Q5	'I would like to see simulation as an analytical tool for process improvement more intensively used in the company.'	X			X
Q6	'I do not prefer VSM to other software process improvement activities that I have participated in earlier.'	X		X	
Q7	'I would like to see VSM used more extensively in the company.'	X		X	
Q8	'The improvements chosen were influenced by the VSM activity.'		X	X	
Q9	'The improvements identified are realistic to implement.'		X	X	
Q10	'The improvements identified will help in improving the software quality.'		X	X	
Q11	'The analytical phase (current map) did not provide valuable input to design with the future state map.'	X		X	
Q12	'The results of the empirical study [59] influenced the choice of improvement actions.'	X		X	
Q13	'The improvement actions identified will be implemented in the future.'		X	X	

Table 2: The aspects evaluated in the questionnaire.

ions about how they perform simulation related activities to support their SE empirical studies. The interview was semi-structured and the questions are available online [60]. The answers from the two researchers were used to compare and reflect on the perception of effort required by researchers in various steps.

As mainly qualitative data was recorded, the analysis of the observation records and notes was initially performed in isolation using open coding technique [61] to highlight and organize the main observed topics. Later, data collected from different sources were used together, making a triangulation to understand the reasons for the findings which are reported in Section 5.3.2.

4.4. Validity threats

Petersen and Gencel [62] reviewed the literature on validity threats based on different scientific schools of thought and recommend a categorization of validity threats encompassing different views. They recommend discussing 1) descriptive validity (factual accuracy), 2) theoretical validity, 3) generalizability (internal and external), 4) interpretive validity, and 5) repeatability.

Descriptive validity is concerned with describing observations made objectively and truthfully.

- There is a threat that the researchers captured information and views provided during the workshops inaccurately. Two actions have been taken to reduce this threat. First, while one of the researchers was having the main moderator role, the remaining researchers were recording the information provided by the participants taking notes. Hence, the data collection did not rely on one individual. Second, the findings and interpretations of the researchers were presented at the beginning of each workshop (member checking).
- We utilized a questionnaire to capture the views of the practitioners, a threat being that questions might be misunderstood. To reduce this threat, the researchers were present while the practitioners filled in the information, so that they could answer questions for clarification.
- The external observer who interviewed the first two authors of the paper, and also acted as an observer during the workshop, was new to the company and their terminology, while the first two authors had some familiarity with the domain and the company.

Theoretical validity is concerned with our ability to capture what we intend to capture, the major threat being confounding factors we are not aware of. It may be

affected by the trust of the practitioners towards the researchers. To reduce the threat, the VSM manager was present during the workshops. Furthermore, the first two authors have a history of working with the company and could refer to previously conducted projects to highlight the trust. Hence, the threat is reduced, though not mitigated.

In the questionnaire (see Table 2) to improve the reliability of the results some questions were phrased negatively. Had a practitioner not responded carefully then we would be able to see a mismatch between their responses to the questions Q6 (“*I do not prefer VSM to other software process improvement activities that I have participated in earlier.*”) and Q7 (“*I would like to see VSM used more extensively in the company.*”). A negative formulation does potentially introduce bias and limits the inferences that can be drawn from the answers. However, we believe that the response to Q-7 compensates for the potential limitation of Q6, where we explicitly ask if they would like “*to see VSM used more extensively*”. Lastly, the consistent responses for the two questions by practitioners add to the confidence in the results indicating that the questionnaire was indeed filled carefully.

- The selection of subjects largely influences the findings. The key stakeholders important for the activity have been identified with the help of the VSM manager. The VSM manager assured that the persons in the workshops have excellent knowledge of the process, and could also assure that actions identified in the workshops are actually implemented.
- Expectations and attitudes towards the VSM activity may affect the results, and hence may alter the results. Overall, given that the practitioners to the largest part evaluated the activity positively, this is not considered a significant threat.

Generalizability is concerned with transferring the results within a context (in this case within the cases) and between different contexts (e.g. other products and companies).

- To assure internal generalizability the VSM manager supported us in the selection of workshop participants. All activities of the overall process were covered, which reduces a threat to internal generalizability.
- We conducted the study in two different cases that were developed at the same company, but followed different processes and both products were of different ages. Product A was a newly developed product, while product B was mature. For product A and

B the baseline situation was very different in terms of availability of data, which had a significant effect on the ability to use simulation. Furthermore, contextual factors may exist that may affect the process of combining simulation with VSM that could not be captured in this study. In order to facilitate the judgment of the generalizability, we provided a detailed account of the context, and the background of the practitioners.

Interpretive validity is concerned with whether the conclusions and inferences made by the researchers based on the data are reasonable or not.

- All three authors have worked with the creation of simulation guidelines [63, 54], and have written synthesis studies on process simulation [64, 43]. The conclusions drawn based on their previous work may affect their interpretation of the findings. In order to reduce this threat, we investigated the results from (a) the views of the practitioners; (b) the researchers conducting the simulation study through the interview conducted by an external observer (third author), and (c) the external researcher acting as an observer during the workshop. These findings could be triangulated. Hence, the threat to interpretive validity is reduced.

Repeatability is supported by describing the research method in detail. Furthermore, we provided a web-page containing the instruments for data collection and the non-calibrated simulation model as the raw data could not be made available due to confidentiality reason.

5. Use and evaluation of the VSM and the simulation support (Contribution 2)

This section reports the second main contribution of the study that focuses on evaluation of simulation-assisted VSM. In this regard, the proposed framework (presented in Section 3) was instantiated at the company and the results are presented in Section 5.1, answering RQ1. Section 5.2 answers RQ2.1 by detailing the contribution of simulation concerned with the wastes and improvements that were identified. Section 5.3 answers RQ2.2, which is concerned with the perception of the participants in the VSM activity.

5.1. Conduct and outcome of the VSM activity at the company (RQ1)

The VSM started with assignment of the following roles and responsibilities to participants of this study:

VSM Manager: In this study, one practitioner took on this role, he had the authority and the power to facilitate and ensure that change happens in the organization. He was also instrumental for conveying the goal and convincing the teams why it was important. He helped in negotiating the appropriate scope and goal for each product. He also ensured that the necessary resources and time was allocated as a priority for VSM workshops.

VSM Facilitators: The first two authors acted as VSM facilitators in this study. Both authors have experiences as facilitators, and the second author has conducted VSM studies earlier.

VSM Participants: In large product development, as is the case in the products being studied here, no one individual will know all the details of the process. Besides, they may feel more strongly about the challenges they face in the work that they are more closely involved with, this may lead to sub-optimization which is contrary to the aim of VSM. There were 5-6 practitioners from each product, with knowledge and experience encompassing the end-to-end process ensuring a variety of perspectives. The number of participants and their median experience in years with the product and the case company is summarized in Table 3. These participants covered all the major phases starting from system management all the way downstream to system testing.

Product	Participants	Experience with the product	Experience with the company
A	5	1.5	11.5
B	6	5.5	18

Table 3: Participants of the VSM workshops

Initiation: Goal for the VSM activity by the upper management was stated as the following: *Reduce the average and variance in end-to-end lead-time by half of its current value without compromising the quality.* Three target aspects highlighted here were the efficiency (measured by average lead-times), predictability (measured by variation in lead-times) and quality (the number of defects detected both internally and by external customers).

For the two products, it was translated as *“Reduce the lead time (on average) from “Start BUC analysis” step to “BUC is ready for release” step by “50%” from currently X “units of time” and the variation in lead-times by “50%” without compromising the quality.”* For the sake of confidentiality we have not reported the actual number or units for the current lead-time.

Once the goal was agreed upon, the product owner made available to us process documentation and measurement data on end-to-end lead-times and any subdivisions of lead-times that were available for the sub-phases. To an extent, this helped the facilitators get familiarized with the overall product development process.

Current process map: This workshop is the first interaction with the participants from the team. So the *“VSM facilitators”* started with introductions followed by what is VSM, how it is done, what are the benefits and what is expected of the participants in the rest of the workshops. The *“VSM Manager”* then presented the goal and the motivation behind it.

Then the participants were asked to *“take one typical BUC through the process from the first to the last step as identified in the goal”*. This way we were able to elicit the information expected in a CSM without forcing a certain notation on the participants. This understanding of the current way of working was fundamental for the next step to identify and understand the *“waste”* in the process.

Waste identification: When identifying waste in the development process using the VSM approach, the following were the main findings:

- **Contradiction between perceived challenges and data:** This was observed when the perceived challenges were compared to the quantitative data analysis results that are typical of VSM for identifying *“wastes”* [65] in the process.
- **Difficulty to estimate process metrics in the absence of data:** Experts find it hard to quantify waiting times and variations in the development process and any critical reflection based on calculations or simulation from such estimates is disregarded as inaccurate.
- **Contradiction between the expected and likely benefits of improvements:** This was possible when proposed improvements were assessed using simulated output with explicit assumptions.
- **General applicability of data analysis and visualizations:** This became evident from the positive feedback we received during the workshops and how these reports are part of the corporate measurement dashboard.

The remainder of this section details how these findings were derived from the data.

Product	Challenges	Votes
A	Code freeze at merge, since all development sprints are in sync	17
	3-week sprints (will not complete something 1.5 weeks earlier Parkinsons law)	16
	Quality of BUCs	15
B	Time spent on initial analysis of BUCs	16
	Large backlog of BUCs waiting to start	16
	Testing-sprint length	6

Table 4: Challenges and prioritization for the two products.

5.1.1. Contradiction between perceived top challenges and data

After reaching a consensus on the accuracy of the CSM, the participants were asked to identify key challenges in achieving the stated goal.

Once a list of challenges was identified, practitioners prioritized them in terms of their potential to contribute towards the stated goal, provided that they are overcome or ameliorated. The top three challenges identified for the products in this study are presented in Table 4. A rudimentary root-cause analysis was performed to understand the causes for these challenges.

Quality related challenges: Quality of BUCs and fault slippage in testing [66] were identified among the initial list of impediments to achieving the stated goal for VSM. However, the team had already adopted “post-mortem” analysis for each sprint analyzing the defect reports and ensuring that their recurrence is avoided. Furthermore, when prioritizing it as a challenge none of the practitioners considered it an outstanding issue. The reasons for the current quality of BUCs were identified and improvements such as early involvement of concerned stakeholders in the definition phase were suggested and implemented.

At this stage, it is interesting to consider that only the expert opinion was captured. Next we did analysis to determine whether the perceived challenges can be backed up with quantitative data.

Lead-times: By clustering the BUCs lead-time by their estimated effort we could identify BUCs that were the best or the worst in their category in terms of how long they took to go through the process. The lead-time of BUCs was grouped according to the estimated effort using different units of measure as used by the two products. The outcome of this analysis for both the products is shown in Figure 5.

Figure 5 shows that the variance in lead-times is omnipresent throughout the size clusters i.e. certain BUCs with similar estimated time for development, take considerably longer (and hence the peaks).

The challenges identified in the first workshop were general (see Table 4) and should affect development times of all BUCs and hence do not explain why a small percentage of BUC (albeit of different sizes) take so much longer than the average case (see Figure 5). Excluding the lead-times of BUCs that took longer than “X” units of time, which means about 30 % of the BUCs, the average lead-time for the remaining BUCs was very close to the target of reducing lead-time by 50%. One hypothesis could be that by finding out the reasons why these 30% of the test cases take longer, and alleviating the reasons behind it we can achieve the target lead-time. And such potential improvement was not expected from addressing the top challenge the experts perceived as major hindrance to reaching the target lead-time.

Waiting times: These are another way to identify inefficiency in a process by analyzing the time that a work-unit is inactive (referred to as waiting time) [9]. The flow of BUCs through the process was analyzed by distinguishing between waiting times and productive times. This was used as input for reflection on the duration of waiting times, the reasons behind them, whether these were acceptable delays, and if not how can these be reduced.

For product A, the waiting times are shown in Figure 6. The data to do this analysis was not available for all the process steps in Product B, however, it was noticed that there is a long waiting time at the start of the development process where an initial analysis has already been done for work that will not be going to start in the near future.

For product A, as shown in Figure 6 almost half of the time, work packages are waiting in backlogs. It also highlights that the artifacts that are idling are becoming obsolete e.g. the preliminary analysis done in case of product B will need to be redone if development starts after a long time. Besides, another threat is that it might not even be relevant by that time. These waiting times within the process also highlight bottlenecks and where the flow is breaking up in the process.

It was again seen that the challenges identified through expert opinion (see Table 4) could not explain the long waiting times in the process. Furthermore, the challenges identified did not have as significant improvement potential as tackling the waiting times offers. Besides, improving the productive time is perhaps much more difficult compared to reducing the waiting times to achieve the specified targets.

Visualizing flow: Waiting times are often correlated with a discontinuous flow of work. Different ways of visualizing the flow are used at the company, e.g. cu-

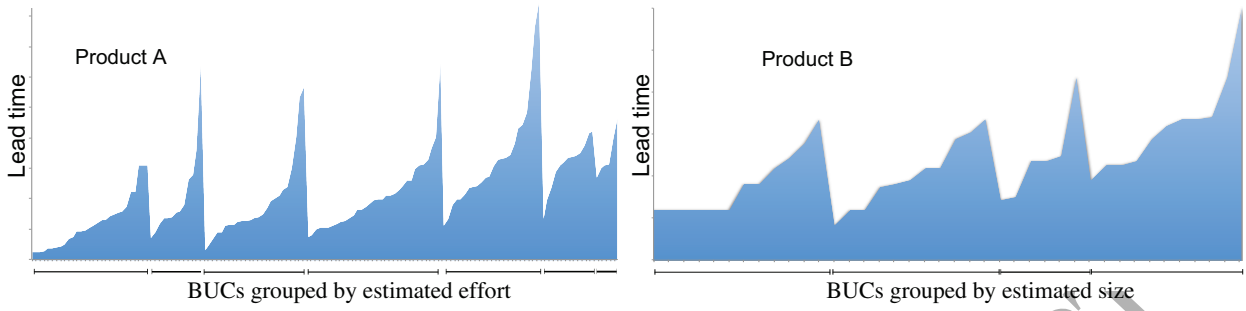


Figure 5: End-to-end lead time for BUCs grouped by their estimated effort.

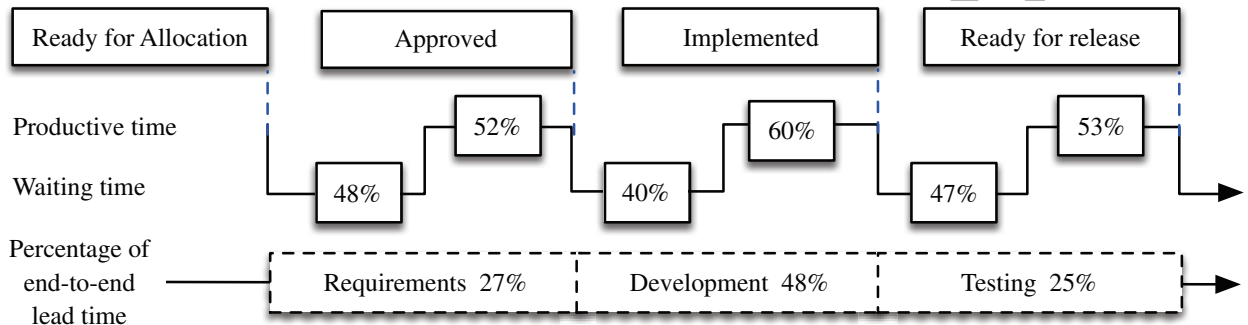


Figure 6: Waiting and productive times for Product A

mulative flow diagrams [47]. In this study, we also used time series analysis (see Figure 7). Figure 7 shows the number of BUCs ready for specific activities (e.g. specification in further detail, or development, etc.). As can be seen in the figure, discontinuity is visible in BUCs “ready for testing” and BUCs “ready for release”. That

is, during a high number of weeks no BUCs are ready. Furthermore, in some weeks a very high number is ready, while in other weeks only one BUC is ready.

The Figure 7 illustrates the pattern of the flow of BUCS through various phases in the time period used for analysis. This means that there were some BUCS

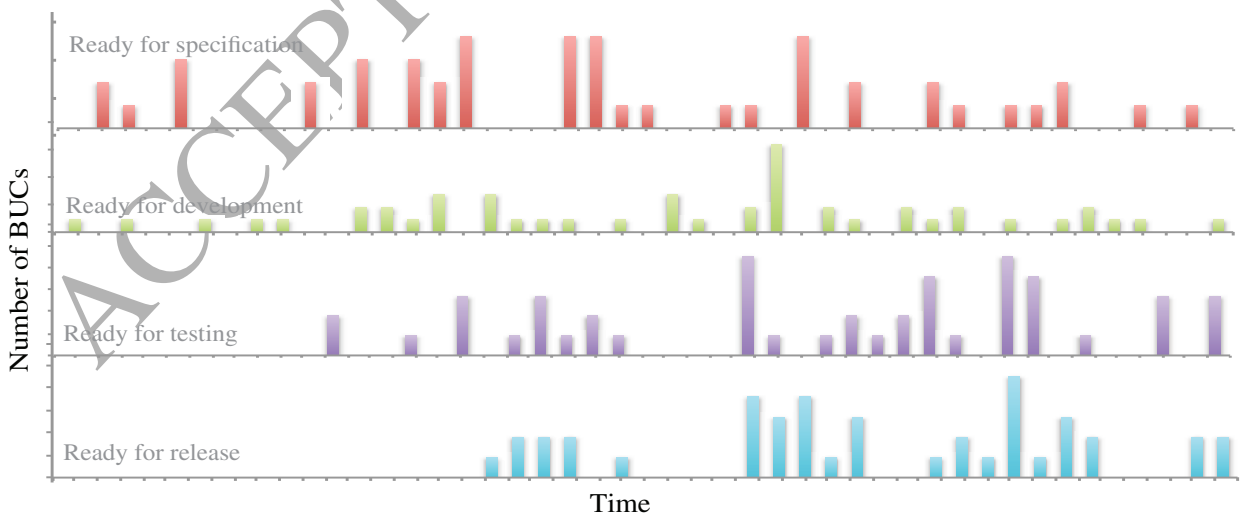


Figure 7: Assessing the flow using a time series plot

that were already specified before the start of the period used for analysis and hence there is a mismatch between what is “ready for specification” and “ready for development”.

The visualization of flow in Figure 7 made it apparent that the perceived improvement to reduce the lead-time by having continuous testing cannot be justified unless the variability in development process is improved. Thus, the additional cost of shortening the test iteration, that requires additional cost in terms of test automation and acquiring more resources for testing, cannot be justified in the existing development flow.

Flow analysis was only conducted for Product A, as Product B did not collect the required data. This also had important implications for the use of simulation. That is, only a simulation model was built based on the data acquired for Product A.

5.1.2. Difficulty to estimate process metrics in the absence of data:

Poppendieck and Poppendieck [3] recommend that by working with people involved in various activities in the development process one should get the average amount of time spent in each activity and also the breakdown of the time into waiting and non-value adding states. In our experience, although we had a sample of representatives for each activity in the process, the participants could estimate the average time it takes for each activity, but they found it impossible to identify how long does an average use case spends in a waiting state. Therefore, this analysis can only be done and have meaningful results if there is an observed dataset to supplement it, as was the case for Product A. In cases of product B the waiting times were not collected by the organization and the participants found it very difficult to estimate general rates. Furthermore, an analysis on top of these estimates was not considered reliable as this was all based on “guess work”.

Some additional interesting observations were not possible using the simulation approach, such as causal influences among variables and the process activities. The main reason are associated to the available data. Since the organization had no goals for simulation before, no data was available for that. This way, it is not possible to discuss reasons explaining the results due to the lack of a mechanism.

5.1.3. General applicability of data analysis and visualizations:

Figure 5 showed similar patterns for both the products although the units for estimation were different. This points to a general applicability of this analysis in

the case company. Also, the list of BUCs identified this way served as a good basis for reflection and validating the preconceived challenges that the team had about their process. The company will soon have it implemented in the dashboard. Already the list of top ten BUCs with the longest lead-times for end-to-end and per phase are available in the dashboard.

The time series analysis (as shown in Figure 7) was found more intuitive by many practitioners to visualize the flow of BUCs and has also become part of the dashboard for the company.

In the case company, for both Product A and B, the utility of tracking the waiting times also became evident. For Product A, where the measurement points were already in use, all the statistical and simulation based analysis was possible. While for Product B and others, there is now a company wide initiative to keep track of some “mandatory” and some recommended measurement points that a team has to collect to keep track of their development process.

Process improvement: In the penultimate workshop, the key findings from all the analysis and reflections in the previous workshops were revisited. For product B, case study results from another Telecommunication company that has transformed from agile to Kanban approach [59] were presented for inspiration. Teams developed an action plan with a list of changes and improvements and drew the FSM.

5.2. Contribution of supporting VSM with simulation (RQ2.1):

In this study, we employed Strategy 4 (see Table 1) to use simulation to assess various alternatives for improvement to derive a future state map. Simulation was used to reason about the likely implications of transitioning towards continuous flow (see Figure 3) instead of the current fixed three weeks iterations. The current way of working (Case 0) and the intended way of working (Case 1) are depicted in Figure 8, which were assessed using simulation. A static representation of the simulation model can be found in [60]. The aim with the intended change (from Case 0 to Case 1) was to reduce lead time, improve quality through early feedback, and to develop the capability of shorter release cycles for the future.

Case 0: Three weeks sprint for development, three week test sprints with hand-offs at the end of the sprint for both development and testing.

Case 1: Three weeks sprint for development, one week test sprint with continuous drops from both development and test.

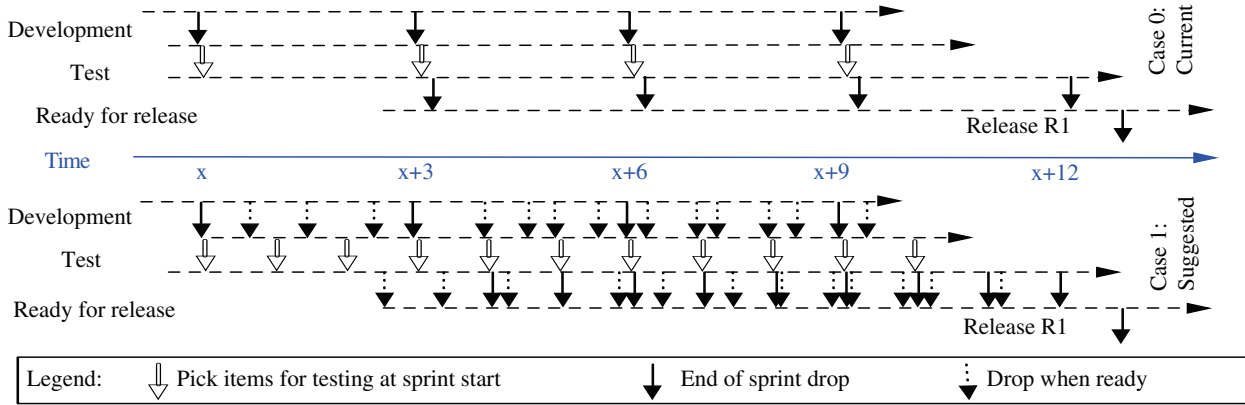


Figure 8: Current development process and suggested development process.

For stochastic simulation, an important step is to identify the random distribution that accurately represents the underlying process behavior. Empirical data from the product was available for the last calendar year, however, the team considered only the data from the last six months as representative of the current process as it had undergone change around that time. This data unfortunately was not sufficient to identify a probability distribution with any statistical confidence. Therefore, for arrival rates of new BUCs, the actual data was used and for development and testing rates normal distribution was used with mean, minimum, maximum and standard deviation as observed in the real data. The reference behavior of the model was checked by determining whether the total cumulative number of BUCs is similar between the simulated case and the actual case, which was true.

Scenarios: Three key scenarios were executed with the help of the simulation model. *Scenario 1* shows results of the two cases with the parameters based on current data. *Scenario 2* shows the outcome of the two cases if the productivity of test and development team stay the same but the variation is reduced (a more predictable process). *Scenario 3* shows the outcome of the two cases above if the waiting times in the process are reduced. The results of the simulation model in terms of cumulative number of BUCs that are ready to be released for the above scenarios can be seen in Figure 9.

From the simulation results (in Figure 9) it was illustrated that the benefits of changing to a shorter test iteration are not very significant. This is because based on the current data there is no continuous delivery from development that a shorter test iteration can leverage from. The result of *Scenario 2* further shows that reducing the variance in the process will lead to more sig-

nificant gains than just the shorter test iteration alone. Lastly, *Scenario 3* shows that an even greater potential exists in reducing the waiting times in the process.

It was identified that often the developers have a tendency to wait for the deadline to commit changes and to update the BUC status. From the discussions based on Scenario 1, the importance was again highlighted for accurate measurements to predict the likely outcome of change.

200 replications of Scenario 1 were done and the outcome was plotted. The variation in the outcome highlighted the unpredictability of the process and the lack of confidence with which we can forecast the impact of the suggested process change. This along with the results of Scenario 2 highlighted the importance of dealing with the variance. Rother and Shook [44] also noted that if the work occurs unevenly overtime (in the form of peaks and valleys) it results in extra burden on people.

Thus, the simulation results showed that the challenges that practitioners perceived as the main impediments to achieving the target of reducing lead-times for product development do not explain the major waiting time and variation in the process data. Moreover, simulation results showed that the benefits expected from addressing the perceived challenges are highly unlikely to deliver the promised results (e.g. due to the current flow of BUCs from development to testing, continuous testing can only be leveraged upon if there is a continuous delivery from the development teams). Furthermore, the high variation in the flow data, became visible in the unpredictability of simulation results thus again emphasizing the importance of more significant challenges that exist in the process and have not been identified by the practitioners.

5.3. Evaluation of simulation-assisted VSM from various perspectives (RQ2.2)

This section reports the results of evaluating both the process and the outcome of conducting simulation-assisted VSM from different key perspectives.

5.3.1. Practitioner's perspective

Usefulness from VSM Manager's point of view: Questionnaire (see Table 2) was used to elicit VSM Managers opinion about the process and outcome of the simulation-assisted VSM. Compared to participants feedback in Figure 10, except Q12, where VSM Manager considered that the results and lessons from a published study did not have influence on the choice of improvement actions. On Q1, Q5 and Q7 he indicated strong agreement and on Q2, Q3, Q4 and Q8 he indicated "moderate agreement". He "strongly disagreed" to the statement Q6 that he does not prefer VSM to other software process improvement. This clearly shows favorable feedback on outcomes of VSM, use of simulation, the process of conducting simulation-assisted VSM.

Usefulness from VSM participants' perspective: The responses of nine practitioners from the two products using a Likert scale are depicted in Figure 10. Please note that Q6 and Q11 were negative questions (therefore semantically it shows positive feedback for VSM process activities for using CSM and VSM in general). The only questions that got negative feedback were Q3 and Q4, where 22% and 20% of the respondents, respectively, were of the opinion that the simulation results were unable to make a convincing case for targeting the "variance" and "waiting time" in the process for improvement. Overall, for questions we got positive feedback with some "strong agreements",

especially on Q9 and Q10 where 50% of the respondents strongly agreed on the realism and likely impact on quality of the improvements and the other 50% moderately agreed to it.

5.3.2. External observer

The external observer noted down his reflections by observing the activities and outcomes of the workshops, and by interviewing the VSM facilitators. His key findings are described below:

Suitability of employed simulation approach: Based on the information and documents given (process descriptions and workshop records), it is possible to observe that problems regarding the product A investigated with simulation resembles a process perspective, evidenced by some characteristics such as synchronization of parallel flows and analysis of waiting times. Probably, it would be easier explored using a discrete-event simulation model, even though such problems can also be handled in system dynamics.

Feasibility of simulation depends on data availability: Although there are VSM-related reasons for adopting simulation as an investigation approach, the data availability was crucial for defining the simulation feasibility. Otherwise, such approach would be prohibitive.

Utility of simulation-assisted VSM: The VSM workshops create a worthy environment for discussing challenges and alternative solutions, from which scenarios (for analysis using simulation) can be determined. These workshops can be potentially used as room for discussing the simulation model, acquire input to determine relevance of variables to reach consensus about what is being simulated, and to validate the simulation model.

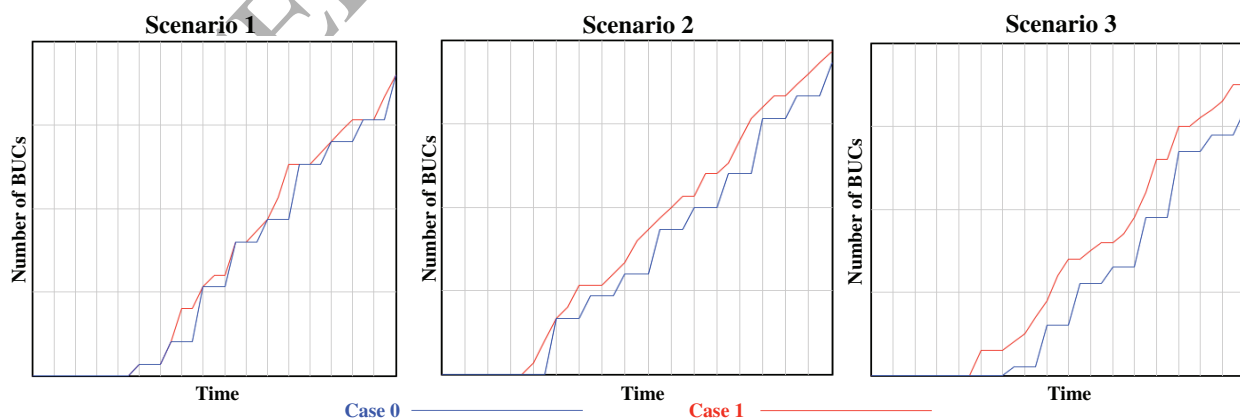


Figure 9: Three scenarios of comparing the current way of working with the suggested changes.

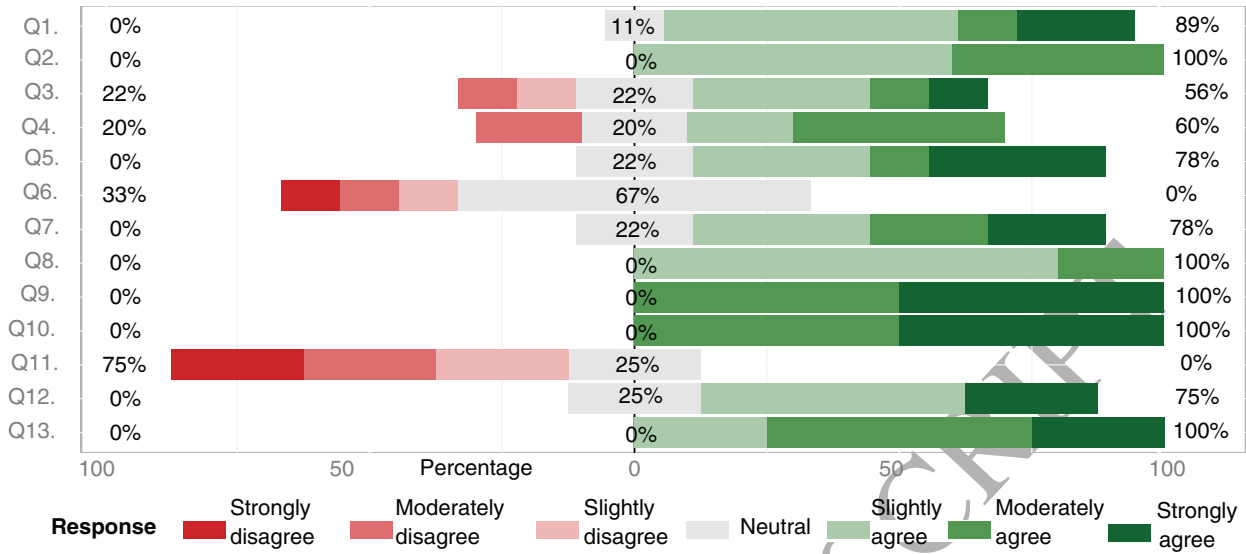


Figure 10: Results of the questionnaire (see Table 2 for questions) to evaluate the simulation-assisted VSM.

When more variables of interest or more scenarios appear during the interactions with the development team, the analysis procedure for simulation outputs will certainly change and needs to be revisited. Therefore, researchers will need to iteratively discuss the results with the team after each revision.

Process for simulation model development and lack of validation: The planning steps for the framework, concentrate on the VSM activities (see Section 3). However, there are no simulation-specific activities or steps in it. Even using the previously discussed strategies [54], important activities of simulation-based studies are not explicit, such as the simulation model assessment and experimental design, and may have been overlooked by the researchers.

The adopted modeling approach in the study followed an iterative process [67]. During the observations, the external researcher could not identify any attempt from the researchers to validate the simulation model with the practitioners.

For the model assessment, the only observation made regards the constant review of the simulation model, when unexpected results started to appear, i.e., when simulation results differ from the actual process behavior. Initially, it appears to be a consequence of the mismatch between the simulation approach (SD) and the problem under investigation, but it also seems to be the selected procedure for model verification.

Besides the resistance shown by one of the teams regarding the statistical analysis of the process metrics (collected in the organization), there was no questioning

about the simulation model validity. Apparently, they assumed the simulation model was correct or, maybe, became overwhelmed by its complexity.

When trying to support VSM by complementing with simulation, the researcher must be aware of the new threats simulation imposes. Even when not having prediction as a goal for the simulation model, its use can still be planned.

Simulation models were not presented upfront: It is important to highlight that, for the two first workshops involving both product A and B, the researchers decided not to present the simulation model. The discussions were based only on the presented simulation results and statistical analysis of data. Only at the third workshop, the researchers presented the simulation model and outputs from the experiments involving the different scenarios. At first, stakeholders appeared a little confused on how to read the model, maybe because of the system dynamics notation. Later, some stakeholders apparently changed their opinion with the simulation tool and its possibility of anticipating scenarios.

Difference in perceived effort spent on simulation modeling: In the interview conducted at the end of the period of observations, it was possible to identify distinct perceptions of effort spent on simulation issues. The researchers have different opinion on which activities they spent more effort: input data analysis or simulation model development. The input data analysis involve the understanding of the company data, using statistic techniques in order to identify or explain certain behaviors. On the other side, the modeling is more

related to how to translate the process specifications (CSM or FSM), along with project data, into the System Dynamic model, which showed to be problematic at the beginning by the mismatch between the SD approach and the kind of problem to be simulated. Such different perceptions might be just a matter of roles played on the study or different kind and level of expertise.

5.4. Retrospective after six months

In October 2014, we returned to the company six months after the conclusion of the VSM activities. This gave us an opportunity to capture information about the impact of the VSM activities on the organization. Two follow-up workshops were carried out to assess the status of the two products in implementing the changes and the adoption of VSM in the company.

The two teams presented their status on each action item and a comparison was done of lead-times, size of backlogs and other measures that were used in the earlier VSM activity. Overall, teams had made significant progress towards implementing the improvements identified through the VSM. At the time of these follow-up workshops there were only slight improvements in the lead-time and variations in the process.

In terms of adoption of VSM, it was found that some reports for doing data analysis, as done in the two case studies reported in the article, have already been implemented in the corporate dashboards, and remaining will be made available in a later release. It is well aligned with the advice in software process improvement literature to ensure automatic collection of data [68].

So far, the practitioner who took on the role of VSM Manager has conducted VSM for six other products.

He stated that he received positive feedback from the teams that were involved in VSM. They considered that the benefits of the workshops outweigh the time spent by participating in them. Even with product teams with skeptical participants the value became apparent once they had participated in the workshops. The dashboard based reports were also useful to reflect objectively on the perceptions that participants had about the state of their product development (e.g. size and place of backlogs). However, the dashboard is not currently used as part of routine work and they will have local “VSM facilitators” to get VSM rolling in the organization.

He considers that the feedback on incorporating simulation has been positive, but there is a need to make it more accessible to practitioners that will take on the role of “VSM facilitators” in the organization. Given that overall one consistent process is followed throughout the organization along with a standard suite of measurements that are collected for each of the products, it

provides some avenues for the use of simulation. The expectation is to explore further the possibility to train the “VSM facilitators” to be able to use the simulation models even though they will not be required to create the models themselves. This would provide them with the tool-set to perform certain analyses, and to highlight that having a mindset to achieve even flow is more important than the traditional emphasis on resource utilization in processes.

6. Discussion

6.1. Evaluating the use of simulation to assist VSM

From the results in Figure 10, to question Q5, 78% of the respondents had a favorable opinion on use of simulation for Software Process Improvement (SPI) activities in the company. Furthermore, all respondents agreed (Q2) that the use of simulation led to more insightful discussions. This reflects positively on the use of simulation to assist VSM.

However, when asked about the two main points that were emphasized through the use of simulation, the importance of reducing “waiting times” and “variation” in the process, with regards to the “waiting times”, in Q4, 60% agreed that simulation helped to make a convincing argument. While on the second point of targeting “variations” in the process, only 56% agreed that the simulation results made a convincing point. Over 40% of the respondents did not agree or were indifferent on the impact of simulation for the two points in the discussion. These respondents were either still unconvinced about the importance of “reducing variance” as a target or they approached the question differently like “simulation was not the sole contributor” in making a convincing point.

The fact that simulation results led to more insightful discussions is enough to justify its use, since the purpose of employing simulation here was to support in decision making and not to make the decisions. However, the cost and time need has to be considered on a case-to-case basis. For this study we approximately (this is a rough estimate as it was done along with other work for VSM between the workshops) spent 10 calendar days on eliciting the scenarios, development and testing of the model, calibration with the data and to analyze and create reports based on the results.

In our previous work with simulation of software processes [54], we modeled the development process at a company. In such cases significant insights and learning can be associated with just the modeling of the process for both the modelers and the practitioners (i.e. regardless of the actual simulation of the model). However, in

the current case the simulation model was solely developed to capture and analyze the scenarios as envisioned by the practitioners.

The key contribution of simulation came from its illustrative power to demonstrate the impact of addressing waiting-times (by comparing the output of reduced waiting times vs. the improvements identified by the practitioners) and variance in the process (multiple runs helped to demonstrate the unpredictability of changes on the process outcome).

At the start of the study, the practitioners were focused on addressing the structure of test iteration. There was an awareness of the challenge and they were already discussing investing in better systems (both software and hardware) to solve the problem. Through VSM based analysis of the flow of BUCs, the waiting times and variance in the process were highlighted. However, the demonstration by the simulation model of the likely benefits of addressing the iteration structure versus investigating the waiting times and the variation in the process helped to make a convincing argument for investigating and reducing the latter in the process.

6.2. *External observer's critique on the validation of simulation models*

The third author observed that not enough effort was spent on validation of the model. We utilized the simulation to evaluate alternative scenarios on how to construct the FSM. In earlier work, we proposed guidelines on how to conduct software process simulation [54]. As part of the process, verification and validation of the model play an important role. However, the degree of verification and validation to be conducted may depend on the purpose of the model usage. In case the model should be used for predictive purposes, and shall be continuously used, a detailed verification and validation should be conducted (including sensitivity analysis, model walk-through, and reference behavior checks).

In this case, the model was checked for correctness by using non random input and analytically calculating the output at each step and the total output from the simulation model. In this study, the simulation model had to capture two scenarios (as shown in Figure 9). First author developed and tested the model, both the second and third author reviewed it and a walk-through of the model was given to the practitioners. In the walk-through we explained how the model represents the two scenarios.

Although data for one calendar year was available, however, the practitioners considered the data from the last six months as representative of the current process. This left us with insufficient data points to identify a

random distribution that can represent the data with statistical confidence. We resorted to using random distribution with minimum, maximum, median and standard deviation based on the actual data. Due to this reason comparison to actual process behavior was not performed. Sensitivity analysis to find critical factors in the process was not done however 200 replications for each scenario were performed and cumulative output was plotted to visualize the dispersion of results.

The intention in this study was to design a “throw-away” simulation to quickly play through alternative scenarios, which meant that only the rough patterns (e.g. there is an effect of waiting time on the possible velocity of the organization) were of interest, rather than saying what the exact gain of velocity is. Similarly, the effect of variations in the process was made visible by the unpredictability of the simulation results in both scenarios when it was run multiple times.

We consider that under the time constraints of the workshops and primarily due to the purpose of the model, the validation done in the study was sufficient for the task. The goal here was not to develop a perfect simulation model but to have a model that makes the assumptions visible and explicit. The model assisted in reasoning about why certain alternatives were better than others.

6.3. *Use of VSM over other SPI activities:*

In response to Q6, only 33% of the respondents favor VSM over other activities and 67% were neutral. However, a positive sign was that none of the respondents indicated that they will not prefer VSM over other SPI. The large majority of 67% could be because the practitioners did not have a comparison point (i.e. they have not participated in any other formal SPI activity before), or in the worst case they are weary of the SPI activities and see this “as good as any other improvement activity” they have participated in. Unfortunately, we do not have the information to ascertain if the respondents have an experience with other SPI activities. Thus, we cannot interpret the “neutrality” of responses here.

33% favorable responses suggest some inclination towards VSM as a SPI activity of choice. Furthermore, responses to Q7 where 78% agreed that they would like to see more extensive use of VSM in the company, supplements this conclusion. The overwhelming positive feedback on the outcome of the VSM activity in the following questions also adds to the confidence in the utility and encourages its further adoption in the software product development.

- 89% of the respondents (in Q1) agreed that they gained new insights about their development processes that they did not have before.
- 100% positive feedback on VSM leading to the identified improvement, the potential to improve the software quality, and that the improvements are realistic and will be implemented in the company (see Q8, Q9, Q10 and Q13).

The second point above also highlights another strength of VSM. Although, the initial drive for improvement came from the top management, but the participants considered the improvements realistic, with the potential to improve quality and their commitment that they will be implemented. This indicates the ability of the VSM activity to create the “*commitment*” which is crucial to the success of any SPI activity [68]. The participants get the end-to-end perspective and they can understand why a certain change is necessary in their process, although they may not be the immediate beneficiaries of the change.

6.4. Other process related considerations:

Application of VSM in SE is relatively new, but the lessons from research on success-factors for software process improvement initiatives are equally applicable in this context. For example: having realistic goals, executive support, sufficient resources, experienced and respected staff, and involving developers as well as managers in the improvement initiative [69].

Most of these factors are inherently encouraged in the bottom-up, non-prescriptive process of VSM, however awareness and conscious consideration of these during the VSM will further improve the likelihood of success.

6.4.1. The role of “VSM Manager”

The importance of having “*change agents*” is well established in software process improvement research [69]. In this study, he was critical to “*negotiate*” a goal for the VSM activity with the product owners, that is both relevant and acceptable (based on the target from the top management). He was also instrumental in ensuring the resources that were required for the activity, which primarily involved practitioners’ availability for the workshops and access to data for the researchers (who took on the role of VSM facilitators).

6.4.2. Terminology

In the initiation meetings before the start of actual VSM workshops, we noticed some reservations on the use of the terms “*value*” and “*waste*” where some product owners considered it too academic. Therefore, in

our communications and in the workshops the emphasis was rather on identifying challenges/impediments in achieving the stated goal, or on identifying the activities and enablers that can help achieve the stated target. Through this wordplay, we managed to avoid a lot of potentially unnecessary discussions without compromising on the aims of VSM where the focus was to improve the end-to-end lead time of the process.

6.4.3. Notation to draw CSM

Contrary to previous experience [8], the respondents in the current study (see Q11 in Figure 10) considered the CSM as useful input in identifying improvement and defining the FSM. This was achieved by not limiting the participants to use any particular notation, which in the previous case was considered too restrictive and simple to capture the actual process [8]. In our experience, giving practitioners the freedom to choose the notation at their will, helped us to focus on what is important i.e. drawing and developing a consensus on how they currently work.

6.4.4. Time taken by VSM

In the guidelines for doing VSM in the manufacturing domain, Rother and Shook [44] recommend that VSM should be a short non-bureaucratic activity where in two days one should have a FSM drawn and ready for implementation, which one may “*fine-tune*” as the progress is made on implementation. However, in this study, it took considerably longer in terms of calendar time to arrive at the FSM, which was due to scheduling challenges from busy schedules for both practitioners and researchers.

Although, the two researchers have a background working with the case company, these products were new for them and it took some time to familiarize with the technical jargon. Also, all the challenges related to data access, extraction and understanding took some time as well owing to dependence on the industry practitioners. Lastly, all the analysis was done manually using spreadsheet and statistics software which required extra processing to clean up the data. The actual workshop time, however, for each product was approx. around 15 hours.

Now that we have practitioners from the case company doing VSMs internally, the challenges that we faced as outsiders will be redundant (the downside however will be the lack of a totally fresh perspective). Furthermore, the reports and visualization that were generated manually to assist in these workshops are now created automatically through a browser-based dashboard therefore that time will also be saved. Theoretically,

it may be feasible to do these workshops in two full days now, but we consider that the workshops should at least be spread on three calendar days. As the workshops were fairly intensive, both for the participants and the facilitators, as often you will have follow-up questions that will require some additional data collection and analysis to answer them.

This will however, still fall well short of the recommended “half hour” threshold for VSM activity set by Kent Beck as quoted by Poppendieck and Poppendieck [3] “it took us about half an hour to come up with this (don’t spend longer than that or you will have too much detail)”.

6.4.5. Waiting times

Interestingly, almost 50% waiting time found in the product development processes here (see Figure 6) is very similar to the typical waiting times found in engineering and manufacturing processes (where work packages were found to be inactive 60% of the times) [9]. The existence of such large waiting times in the software development processes indicates a similar opportunity for improvement in the SE context as found in other domains.

6.5. Adoption of the proposed analyses and the VSM process guidelines at the case company

We have been involved in two subsequent cases of VSM at the case company as well. In one of these, a practitioner from the case company (who was involved in the VSM for product A and B) shared the VSM facilitator role and in the latest instance, she took over the entire responsibility and the researchers were involved in a supportive role (doing the data analysis in the background). The practitioner is also involved in training other employees in sites in Germany, China, India and the US to facilitate local teams to start adopting VSM at their end-to-end processes.

As mentioned in the previous section, all the reports of quantitative data analysis on lead times and flow are now part of the VSM dashboard to facilitate the analysis of their development processes. There is also an active education drive explaining and motivating teams to report the new measurements, use the dashboards and conduct VSM.

The resulting processes in the two cases reported in this paper have not adopted a pull-oriented approach, but with the new measurements and visualizations in place the teams can better quantify the backlog sizes, waiting times and variations in the process output. Thus, this should be the necessary impetus for the teams to transition to a pull-oriented approach in the future.

7. Conclusion

This study reports the strategies, the process for assisting VSM with simulation, and the results of its application in two industrial cases.

RQ1: *Is the value stream mapping process adapted from Khurum et al. [8] capable of establishing the current state, identifying waste, and developing an improved future state of the process?* The adaptation of VSM for the SE context [8] was found useful to provide a systematic approach for establishing the current state map, identifying challenges and establishing a desired future state. Practitioners expressed satisfaction with each of these outcomes i.e. the current state map sufficiently captured the development process and provided useful input for process analysis, the challenges and the corresponding improvements were attributed to the VSM activity, are considered realistic to implement and are expected to improve the quality of the products. Overall, practitioners had a positive experience with the use of VSM and want to continue the use of VSM for continuous process improvement.

RQ2: *How useful is the combination of simulation and value stream mapping, and what are its strengths and limitations in that regard in the software engineering context?* Based on literature, five strategies to support VSM with simulation were identified. We see that there is a potential for making opportunistic use of available data (i.e. where the existing measurement program collecting data in an organization can fulfill the needs of the simulation model) to develop throw-away simulation models for all the strategies. However, Strategy 5 in our opinion is too restrictive in its need for the use of a notation for VSM that can be automatically translated to a simulation model. We contend that the use of simulation should be argued and justified on a case-to-case basis, whether we have a problem that can be explored with the use of simulation given the cost of simulation (time taken to develop, verify and analyze the simulation results), accuracy and reliability of results (subject to availability and quality of data) and the time frame of the VSM process.

Models without supporting data are no more than a visual metaphor [70]. However, a lack of empirical data is a common challenge faced by software process simulation modelers [10]. In this study, simulation was used as a tool for reflection and reasoning about changes in the development process. It was clear that simulation results led to more insightful discussions in both cases for product A and B. However, for Product B (due to the unavailability of data) data from the Product A was used for the calibration of the model. In this case, prac-

tioners from the latter product argued that the results were less applicable for their case.

In other cases where practitioners generally have a dismissive attitude of “*not applicable here*” towards empirical evidence [71], it is likely that such practitioners will dismiss the results of simulation altogether if it were not calibrated with their data. The likelihood of such reactions is even higher if the simulation results do not align with their perception. Thus, a lack of data threatens the usefulness of SPSM even as a tool for reflection.

RQ2.1: Which wastes and improvements could be identified with VSM combined with simulation? In this study, we employed “*Strategy 4*” (see Section 3) i.e. to use simulation to support the reasoning and choice of alternatives for improvement to derive the future state map for the products with some success (based on the feedback of practitioners). Unanimously, practitioners indicated that the use of simulation led to more insightful discussions.

Simulation contributed substantially in highlighting the potential of reducing waiting times and improving the flow in the process. The ability to model uncertainty in the process was visibly the most beneficial in providing a realistic test bed to assess the improvements by making the assumptions of practitioners explicit and providing input for choosing between alternatives. However, based on their feedback, practitioners will not solely rely on the results of the simulation model.

RQ2.2: How do practitioners, facilitators and an external simulation expert perceive the process and outcomes for the VSM assisted with simulation?

Simulation-supported VSM certainly gained practitioners’ confidence in the improvements. They considered them more realistic and thus considered them more likely to be implemented. Satisfactory progress towards implementing the improvements was observed in the follow-up workshops six months after the VSM activity.

Furthermore, based on the feedback from the practitioners, we are less likely to see a transfer of technology when it comes to software process simulation. There is keen interest in the simulation models for supporting VSM and in other avenues where they will like to use simulation models however the interest at least in the case company stopped at the use of the models. As shown in these cases, simulation can be useful, however it is likely to remain a tool for the use of researchers and consultants who can identify appropriate problems where the cost/time for the use of simulation can be justified.

Positive feedback on the VSM process that was followed in these two cases, the participants’ satisfaction with the outcomes of the study, impact on the case organization in the form of new reports in the dashboard and the adoption and training of other sites to utilize VSM, and further application of VSM on six other products indicates a positive outcome/impact of the studies reported in this article.

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